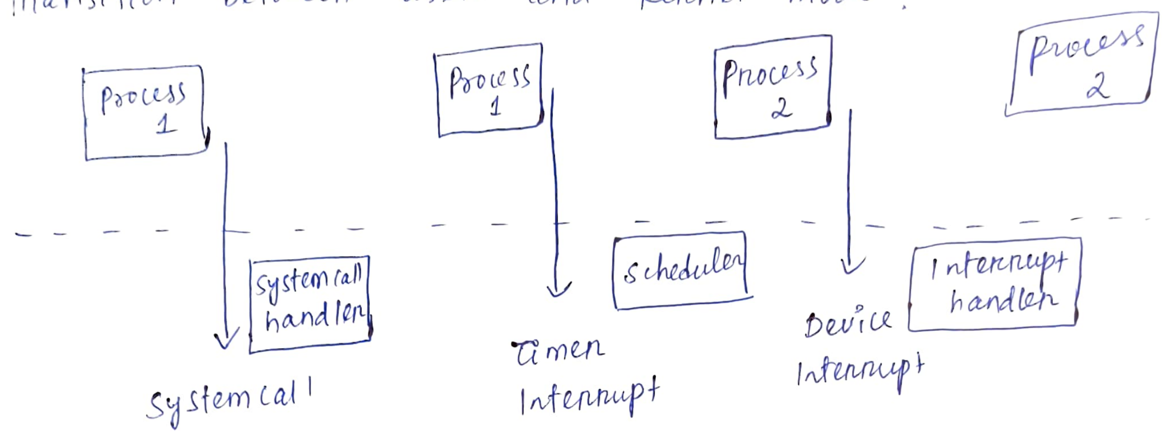
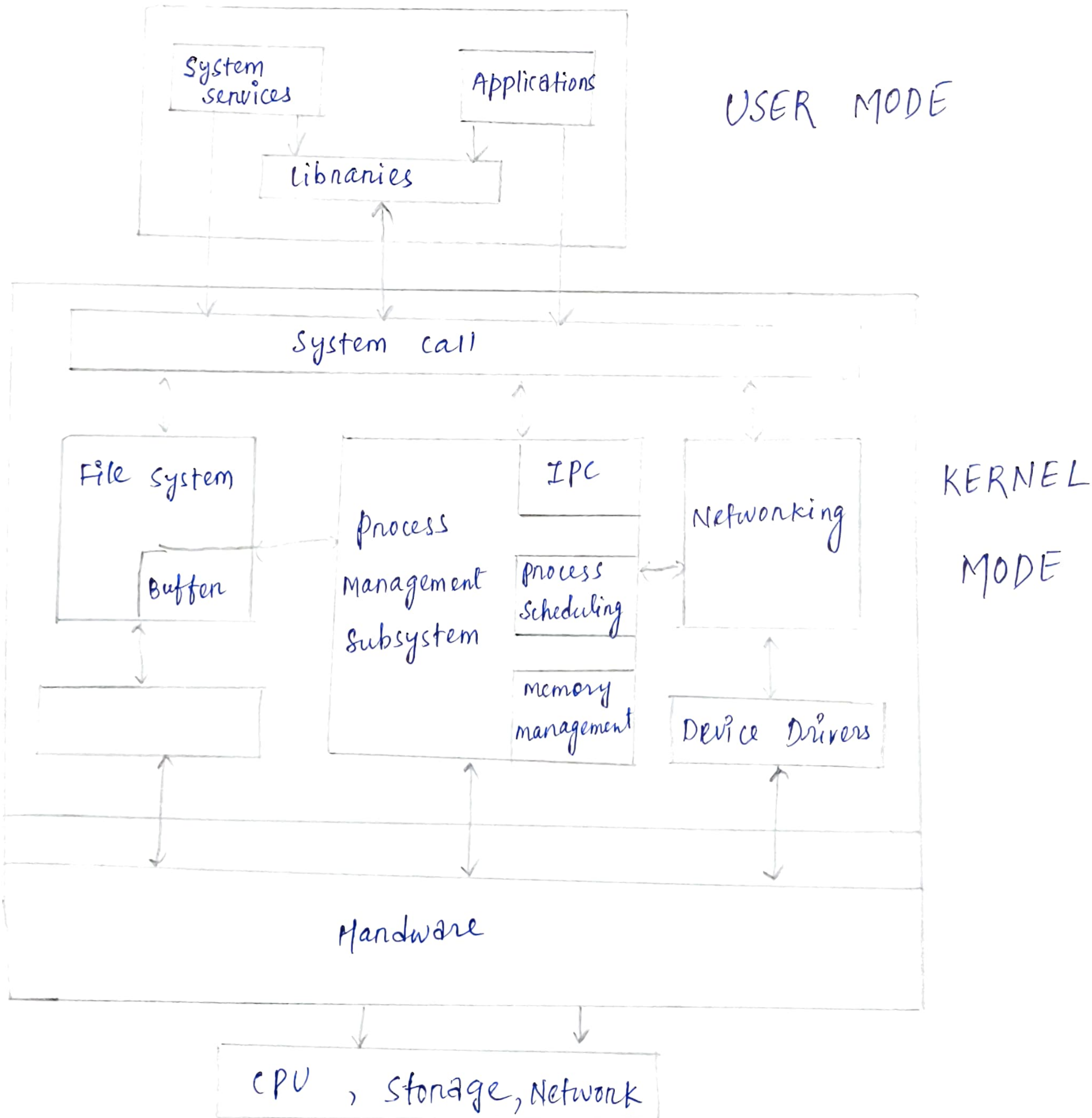
Task1:

Linux Kernel Architecture:

Transition between user and kernel mode:





Major component of kernel Architecture :

- The process scheduler
- File system
- Memory management
- Device Drivers
- system call Interface
- The Networking unit
- Inter - process communication

The Process scheduler :

- kernel is responsible for fairly distributing the CPU time among all the processes running on the system
- It ensures multitasking by determining which process runs next.

File system

- Manages networking protocols and connections providing a disis for ac disks.
- Handles the organisation, storage, retrieval, and manipulation of data on storage devices.
- It works as a Interface to access stored data across different filesystems and physical storage media

Memory management:

- Allocate and deallocates memory space to processes
- It manages physical and virtual memory, including paging and swapping.

Device Drivers

- It is interface between the hardware and the kernel.
- Each device driver controls a specific hardware component like a disk, keyboard or network card.

System call Interface

- provides an entry point for user applications to interact with kernel services.
- converts user-level requests to kernel-level operations.

Networking Unit

- Manages data communication between devices over networks.
- Implements standard protocols like TCP/IP

Inter-process communication (IPC)

- It refers to the mechanism provided by the OS to allow processes to communicate with each other and synchronize their actions.
- The key IPC mechanisms are pipes, FIFOs, message Queues, shared memory, semaphores, Sockets.